# Surrogate Inverse-Variance Weighting for Intervention Harvesting on Heavy-Tailed Click Logs

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## Abstract

Intervention harvesting [1] estimates the position-bias ratio $p_k/p_{k'}$ between positions $k$ and $k'$ by aggregating, across many $(\text{query}, \text{doc})$ cells indexed by $i$, the per-cell click-rate ratio at the two positions. Let $N_k^i$ denote the impression count of cell $i$ at position $k$. On real click logs these counts are extremely heavy-tailed: on the full KDD Cup 2012 Track 2 log, 60% of interventional $(q, \text{ad})$ pairs have $\le 5$ total impressions while 0.1% of pairs hold 42% of all impressions. The way cells are weighted is therefore decisive. The intervention-harvesting estimator of Agarwal et al. uses *uniform* per-cell weighting; we propose two new count-only alternatives—a simple *min-count* weight $\omega_i = \min(N_k^i, N_{k'}^i)$ and a *harmonic* weight $\omega_i = N_k^i N_{k'}^i/(N_k^i + N_{k'}^i)$. *Why harmonic?* Under the Position-Based Model (PBM), the per-impression click variance is $p_k r_i(1-p_k r_i)/N_k^i$, which depends on the unknown per-cell relevance $r_i$. Replacing the $r_i$-dependent factor with a constant yields a *count-only surrogate* of the Bernoulli variance that is exact when the per-cell relevance $r_i$ is constant across cells, and a controlled approximation otherwise. Among count-only weights, harmonic is the inverse-variance optimum on this surrogate, and a Cauchy–Schwarz inequality guarantees it reduces the surrogate for any impression counts. The full inverse-variance optimum would require knowing $r_i$ per cell and is infeasible on heavy-tailed data. We assess finite-sample efficiency at fixed weights via the closed-form delta-method asymptotic variance of the weighted ratio estimator [10, 12]. On the full KDD log, harmonic's asymptotic relative efficiency over uniform weighting is 4.5× and remains robust to weight capping. Under PBM mis-specification, different weight schemes target different weighted averages of cell-specific ratios—an estimand caveat we make explicit. Synthetic and Yahoo-LTR semi-synthetic experiments with ground truth corroborate the efficiency prediction (up to 54% variance reduction at high imbalance); min and harmonic are empirically comparable, and we recommend harmonic for its unconditional theoretical guarantee.



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## 1 Introduction

Position bias systematically corrupts click signals used for learning-to-rank [7]; intervention harvesting [1] estimates position-bias ratios $R_{k,k'} = p_k/p_{k'}$ from naturally occurring ranking variations without explicit randomisation. On production click logs, the per-cell counts that feed this estimator are extraordinarily heavy-tailed. On the full KDD Cup 2012 Track 2 release (235M impressions; 4.6M interventional pairs at adjacent positions $(1, 2)$), **60.7% of pairs have $N_1 + N_2 \le 5$ while 0.1% of pairs carry 42% of all impressions** [8]. Public benchmarks can hide this shape: the released Baidu-ULTR [15] aggregation has 94% of $(q, d, \text{pos})$ cells with $N = 1$ and 78% of adjacent-pair impression ratios within $[0.5, 2]$, which we attribute to the per-cell deduplication used in the public release rather than to the underlying production logs (the unaggregated KDD release, by contrast, preserves the heavy-tailed production distribution).

Equal-weighting heteroscedastic estimators is statistically sub-optimal by classical inverse-variance theory [4], a result the meta-analysis literature has championed for decades. The intervention-harvesting estimator of [1] uses uniform per-cell weighting (each cell's click rate enters the sum with weight 1). **We propose two new count-aware alternatives**: the *min-count* weight $\omega_i = \min(N_k^i, N_{k'}^i)$ and the *harmonic* weight $\omega_i = N_k^i N_{k'}^i/(N_k^i + N_{k'}^i)$. Empirically, min and harmonic perform comparably on KDD (Section 4); the harmonic proposal is preferable because of its stronger theoretical foundation: it is the inverse-variance optimum of a relevance-agnostic surrogate (Theorem 2) and the only one of the two with an unconditional variance-reduction guarantee against uniform (Theorem 3). Existing variance-reduction arguments on heavy-tailed real logs have relied on *bootstrap* resampling, which we show can be misleading for inverse-variance-style weights on singleton-dominated data (Section 3).

**Contributions.** **(1)** We show that the harmonic weight is the inverse-variance optimum on a *relevance-agnostic count-only surrogate* of the per-cell variance under PBM (Theorem 2), and prove a Cauchy–Schwarz inequality showing it reduces this surrogate for any impression counts (Theorem 3). The exact inverse-variance oracle for the more general asymmetric surrogate is also characterised, and harmonic is its parameter-free special case (Theorem 4). **(2)** We adopt the textbook closed-form asymptotic variance of the weighted ratio estimator as a finite-sample diagnostic that is properly aligned with conditional-on-counts efficiency on heavy-tailed data, where naive pair-bootstrap is misleading. **(3)** On the full KDD log, harmonic delivers an asymptotic relative efficiency of 4.53 over uniform on this surrogate – meaning uniform weighting would need roughly 4.5× the data to match harmonic's precision – with the plug-in agreeing within 1–2% of empirical click-resample standard error and remaining robust to weight capping. A Cochran-$Q$ heterogeneity test ($Q/\text{df} = 3.83$) confirms substantial PBM mis-specification, under which the choice of weight is a precision/estimand trade-off rather than a search for a single "true" position-bias ratio. Synthetic and Yahoo-LTR experiments with ground truth corroborate.

## 2 Estimator and Theory

*Notation.* Throughout, $q$ is a query, $d$ a document/ad, $k$ a result position, and $i$ indexes the $(q, d)$ cells in the interventional set. The Position-Based Model (PBM) assumes $P(\text{click} \mid q, d, k) = p_k r_{q,d}$ with $p_k \in (0, 1]$ the position-$k$ examination probability and $r_{q,d} \in [0, 1]$ the $(q, d)$ relevance; we abbreviate the per-cell relevance as $r_i$ and write the true per-cell click probability as $\theta^i_k = p_k r_i$. We observe $N^i_k$ impressions and $C^i_k$ clicks for cell $i$ at position $k$, with empirical rate $\hat{\theta}^i_k = C^i_k / N^i_k$. Our target is the position-bias ratio $R_{k,k'} = p_k / p_{k'}$. Per-cell weights are $\omega_i \geq 0$. Given the set of $(q, d)$ pairs that appeared at both positions $k$ and $k'$, the estimator of [1] for $p_k / p_{k'}$ is

$$\hat{R}_{k,k'} = \frac{A}{B} = \frac{\sum_i \omega_i \, \hat{\theta}^i_k}{\sum_i \omega_i \, \hat{\theta}^i_{k'}}, \quad \hat{\theta}^i_k = \frac{C^i_k}{N^i_k}. \tag{1}$$

$\hat{R}_{k,k'}$ is the standard *self-normalised weighted ratio* estimator: a sum of weighted per-cell click rates divided by another such sum. We say a weight scheme is *ratio-unbiased* when, under PBM, the ratio of the conditional expectations of the numerator and denominator equals $p_k / p_{k'}$ – note this is not the same as $\mathbb{E}[\hat{R}] = p_k / p_{k'}$, because expectation does not commute with ratios; the finite-sample bias of $\hat{R}$ is $O(1/N)$ under standard regularity. Under PBM, every count-only weight $\omega_i = \omega(N^i_k, N^i_{k'})$ satisfies this property, and $\hat{R}$ is consistent for $p_k / p_{k'}$ [1].

Proposition 1 (Conditional delta-method variance). *Given the vector of impression counts* $\mathbf{N} = \{N^i_k\}$, *the delta method [12] gives*

$$\mathrm{Var}(\hat{R}_{k,k'} \mid \mathbf{N}) \;\approx\; R^2_{k,k'} \left[ \alpha \, V_k(\omega) + \beta \, V_{k'}(\omega) \right],$$

*with* $V_r(\omega) = \sum_i \omega_i^2 / N^i_r / (\sum_i \omega_i)^2$, $\alpha = (1 - p_k)/p_k$, *and* $\beta = (1 - p_{k'})/p_{k'}$.

*Relevance-agnostic count-only IVW..* The exact Bernoulli variance is $\mathrm{Var}(\hat{\theta}^i_k) = p_k r_i (1 - p_k r_i)/N^i_k$, which depends on the cell relevance $r_i$. Plug-in inverse-variance weights with $\hat{r}_i$ estimated per cell define the *θ-aware oracle* of Theorem 4; in heavy-tailed click logs $\hat{r}_i$ is itself extremely noisy on the $\sim 60\%$ singleton mass, making this oracle infeasible in practice (Table 2: ARE = 0.002). We therefore work with a *relevance-agnostic surrogate* that drops the $r_i$-dependent factor from the cell-level variance; this surrogate is exact when the per-cell relevance $r_i$ is constant across cells, and a controlled approximation otherwise. The inverse-variance optimum [4] on this surrogate, restricted to count-only weights, is the harmonic mean of the two cell counts. We make this precise:

Theorem 2 (Count-only IVW on the surrogate). *Among all non-negative weights* $\omega_i = \omega(N^i_k, N^i_{k'})$, *harmonic weighting minimises the count-only delta-method variance surrogate* $V_k(\omega) + V_{k'}(\omega)$.

Theorem 3 (Surrogate guarantee, unconditional in counts). *For any impression counts* $\{N^i_k, N^i_{k'}\}_{i=1}^n$, $V_k(\omega^h) + V_{k'}(\omega^h) \leq V_k(\mathbf{1}) + V_{k'}(\mathbf{1})$. *The guarantee is unconditional in the counts but conditional on* $V_k + V_{k'}$ *as the objective; it implies improved* $\mathrm{Var}(\hat{R} \mid \mathbf{N})$ *via Proposition 1 when* $\alpha \approx \beta$. *For the asymmetric* $\alpha V_k + \beta V_{k'}$ *surrogate the bound is empirical only, but we verify on full KDD that harmonic reduces it by essentially the same factor as the symmetric one (Section 4).*

Sketch. Substituting $\omega^h_i = N^i_k N^i_{k'} / (N^i_k + N^i_{k'})$ collapses the LHS to $1/W$ with $W = \sum_i \omega^h_i$. The RHS is $(1/n^2) \sum_i 1/\omega^h_i$. Cauchy–Schwarz gives $n^2 \leq W \sum_i 1/\omega^h_i$. □

Theorem 4 (Exact asymmetric oracle). $\alpha V_k(\omega) + \beta V_{k'}(\omega) \geq (\sum_i 1/s_i)^{-1}$ *with* $s_i = \alpha / N^i_k + \beta / N^i_{k'}$, *attained at* $\omega^\star_i \propto N^i_k N^i_{k'} / (\alpha N^i_{k'} + \beta N^i_k)$. *Harmonic is the* $\alpha = \beta$ *special case.*

Theorem 3 is unconditional in the counts – a property that distinguishes harmonic from the min-count variant, which has no such guarantee. Theorem 4 provides the oracle benchmark; in synthetic experiments substituting the true $\alpha, \beta$ into $\omega^\star$ yields variance within ±2% of harmonic. Complete proofs and additional sensitivity analyses are provided in the supplementary material.

*Limits of the symmetric guarantee.* The unconditional bound in Theorem 3 concerns $V_k + V_{k'}$. The asymmetric analogue $\alpha V_k(\omega^h) + \beta V_{k'}(\omega^h) \leq \alpha V_k(\mathbf{1}) + \beta V_{k'}(\mathbf{1})$ does *not* hold unconditionally: a closed-form $n = 2$ counterexample with $(N^1_k, N^1_{k'}) = (1, 100)$, $(N^2_k, N^2_{k'}) = (100, 1)$ and $(\alpha, \beta) = (0.01, 1)$ has harmonic strictly worse than uniform on the asymmetric surrogate. The configurations under which this fails are degenerate (each cell informative at exactly one position); we have not encountered them in any realistic count distribution we tested. The $\omega^\star$ in Theorem 4 remains the exact oracle for all $(\alpha, \beta)$.

## 3 Plug-in Asymptotic Variance as Diagnostic

*$\widetilde{V}$ in closed form.* The delta method applied to $\hat{R}_{k,k'} = A/B$ gives

$$\widetilde{V}(\omega) = \frac{\sum_i \omega_i^2 \, \theta^i_k (1 - \theta^i_k)/N^i_k}{A^2} + \frac{\sum_i \omega_i^2 \, \theta^i_{k'} (1 - \theta^i_{k'})/N^i_{k'}}{B^2}, \tag{2}$$

where $A = \sum_i \omega_i \theta^i_k$, $B = \sum_i \omega_i \theta^i_{k'}$, and clicks at the two positions are conditionally independent given counts. The standard error of $\hat{R}_{k,k'}$ is then $\mathrm{SE}(\hat{R}_{k,k'}) = |\hat{R}_{k,k'}| \sqrt{\widetilde{V}(\omega)}$. Plugging in shrinkage estimates $\hat{\theta}^i_k = (C^i_k + \bar{\theta}_k)/(N^i_k + 1)$ with $\bar{\theta}_k$ the marginal position-$k$ CTR yields the textbook self-normalised IS variance estimator [9–11].

*ARE..* For schemes $a, b$, $\mathrm{ARE}(a, b) = \widetilde{V}(\omega_b)/\widetilde{V}(\omega_a)$: scheme $b$ needs $\mathrm{ARE}(a, b)$ times the data of $a$ for equal precision. $\widetilde{V}$ scales as $1/\mathrm{ESS}(\omega)$, with Kong's effective sample size [9] $\mathrm{ESS}(\omega) = (\sum_i \omega_i)^2 / \sum_i \omega_i^2$.

*Why pair-bootstrap is misaligned here.* A naive bootstrap that resamples $(q, d)$ pairs estimates a different quantity than the conditional-on-counts efficiency we care about, for three reasons. *(i)* Averaging over $\sim 60\%$ near-singleton pairs under uniform produces a smooth resampling distribution whose tightness reflects averaging-of-noise rather than estimator precision. *(ii)* Pair resampling perturbs the high-information cells that drive any inverse-variance estimator, so concentrated weight schemes are systematically penalised by the resampling design, not by their statistical behaviour. *(iii)* Pair-bootstrap entangles estimator efficiency with model fit; $\widetilde{V}$ separates them (Section 4). The appropriate empirical baseline against which to validate the plug-in is the click-resample

**Table 1: Plug-in SE vs. click-resample SE across KDD subsamples, and ARE(harmonic, uniform) at each scale. Click resample only computed at small $f$ for runtime; agreement is within 1–2% where measured. ARE grows with data volume because heavy-tail mass becomes more pronounced at larger scales.**

| $f$ | pairs | $\widehat{\mathrm{SE}}_U^{\mathrm{plug}}$ | $\widehat{\mathrm{SE}}_U^{\mathrm{boot}}$ | $\widehat{\mathrm{SE}}_H^{\mathrm{plug}}$ | ARE |
|---|---|---|---|---|---|
| $10^{-3}$ | 4.3K | 0.131 | 0.132 | 0.111 | 1.66 |
| $10^{-2}$ | 54K | 0.039 | 0.038 | 0.028 | 2.21 |
| $10^{-1}$ | 542K | 0.012 | – | 0.007 | 3.07 |
| 1.0 | 4.61M | 0.0042 | – | 0.0020 | **4.53** |

**Table 2: Plug-in delta-method asymptotic variance $\widetilde{V}$, SE of $\hat{R}$, ARE vs. uniform, and ESS on full KDD Cup 2012 Track 2 ($n_{\text{pairs}} = 4.61$M). Harmonic, min, and the count-only oracle all reach ARE $\approx 4.5$. The full-$\theta$ oracle is pathological because per-cell $\hat{\theta}_i$ is too noisy at this scale.**

| Scheme | $\hat{R}$ | SE | ARE | ESS |
|---|---|---|---|---|
| Uniform | 1.683 | 4.24e-3 | 1.00 | 4.61M |
| Min | 1.712 | 2.05e-3 | 4.41 | 2,526 |
| **Harmonic** | 1.710 | 2.02e-3 | **4.53** | 1,849 |
| Count-oracle (0.5,1) | 1.715 | 2.02e-3 | 4.57 | 1,760 |
| $\theta$-aware oracle | 2.343 | 0.148 | 0.002 | 1.1 |

(parametric Bernoulli bootstrap conditional on counts), which we adopt for calibration.

*Calibration and stability at scale.* Table 1 reports plug-in SE versus empirical click-resample SE across KDD sample fractions, together with ARE(harmonic, uniform) at each scale. Because both $\widetilde{V}$ and click-resample SE share the conditional Bernoulli + cross-cell-independence assumption, their agreement is primarily an implementation-correctness check; a stronger probe of the independence assumption would be a query-cluster bootstrap, which is sensitive to Cochran-$Q$-style heterogeneity. We report pair-bootstrap numbers in supplementary material. The ARE *grows* with sample size ($1.66 \to 4.53$) because the heavy-tailed structure of the count distribution becomes more pronounced as more data is included rather than averaged out.

## 4 Real-Log Results: KDD Cup 2012 Track 2

*Data.* Full release: 149.6M raw rows expanding to 235.6M impressions, 63.8M unique $(q, \text{ad}, \text{pos})$ cells, 8.2M clicks (CTR 3.49%). Pair $(1, 2)$ interventional set: 4.61M $(q, \text{ad})$ pairs, 132.3M impressions, median pair size 4, max 2.47M; 60.7% of pairs have $N_1 + N_2 \le 5$.

*Headline (Table 2).* ARE(harmonic, uniform) = 4.53: **uniform requires $\sim 4.5\times$ the impressions of harmonic for matching precision** on the relevance-agnostic surrogate. The min-count proposal achieves 4.41 on the same surrogate (−2.7% relative): harmonic and min are essentially tied empirically on KDD, and we recommend harmonic because it carries the unconditional guarantee (Theorem 3) and the parameter-free derivation, not because of a large empirical margin. The count-only oracle (4.57) is the ceiling within the surrogate. The $\theta$-aware oracle, which would be optimal under classical inverse-variance theory with known $r_i$, plugs in $\hat{r}_i$ that is too noisy on the singleton-dominated tail and collapses to ARE = 0.002: a substantive finding rather than a side note—it shows that count-only weighting is robust precisely where plug-in IVW fails.

**Table 3: Stratified ARE(harmonic, uniform) by $N_1 + N_2$ tertile on full KDD (the bottom quartile boundary collapses into the lowest tertile because singletons dominate). Schemes are tied on the noise-dominated strata; harmonic dominates $\sim 5\times$ on the high-information stratum where the actual estimation signal lives.**

| $N_1 + N_2$ tertile | $n$ pairs | $\widetilde{V}_U$ | ARE$(h, u)$ |
|---|---|---|---|
| Lowest | 1.99M | 2.3e-5 | 1.02 |
| Middle-low | 1.38M | 1.9e-5 | 1.09 |
| **Highest** | 1.25M | 7.6e-6 | **4.98** |

*Asymmetric surrogate (empirical).* With $\hat{p}_1 = 0.0501$, $\hat{p}_2 = 0.0256$ estimated from the marginal CTRs, $\alpha = 18.97$, $\beta = 38.08$, $\alpha/\beta \approx 0.50$. Harmonic reduces $\alpha V_k + \beta V_{k'}$ by a factor of 4.75 versus uniform. Both this number and the symmetric reduction of 4.84 are deterministic functionals of the observed counts—the gap reflects the intrinsic asymmetry of $(\alpha, \beta)$, not Monte-Carlo noise—and the two are within 2% of each other. The closed-form asymmetric counterexample of Section 2 requires strongly *anti-correlated* $(N_k, N_{k'})$ across cells (each cell informative at exactly one position); on KDD the per-pair counts are positively correlated, as items popular at position 1 are typically also popular at position 2, and we conjecture this is generic for production logs.

*Stratified ARE (Table 3).* Schemes tie on noise singletons (where there is nothing to estimate) and harmonic dominates uniform by $5\times$ on the high-information stratum. The gain comes from *not* treating singletons as equal to million-impression cells.

*Weight-cap robustness.* The harmonic ESS is 1,849 out of 4.61M pairs: concentration on a small subset of high-count cells is what produces the surrogate $\widetilde{V}$ reduction, and the trade-off is sensitivity to those cells, which we probe directly by capping. Capping $\omega_i^h$ at the 99.99th percentile (threshold 2,079; 462 pairs capped) lifts ESS to $5.2\times10^4$ and still gives ARE = 4.12; at the 99th percentile, ARE = 2.83 (ESS $7.6\times10^5$); even at the 90th, ARE = 1.86. Harmonic beats uniform robustly across capping levels.

*Comparison of count-only weighting schemes.* Table 4 compares harmonic to the natural alternatives on full KDD. Among count-only weights, harmonic is best on $\widetilde{V}$. Log-counting is too flat (treats singletons nearly as well as million-impression cells); the geometric mean and min-count concentrate more aggressively but neither matches the inverse-variance optimum. Capping at the 99th percentile preserves a $2.83\times$ advantage but shifts the estimand ($\hat{R} : 1.71 \to 1.61$), consistent with the estimand discussion below.

*Estimand under PBM mis-specification.* PBM is a known approximation; on real logs different weight schemes estimate different

**Table 4: Count-only weighting schemes compared on full KDD via plug-in $\widetilde{V}$. Schemes that concentrate more aggressively give lower $\widetilde{V}$ but smaller ESS. Harmonic is the inverse-variance optimum among count-only schemes (Theorem 2).**

| Scheme | $\hat{R}$ | SE | ARE | ESS |
|---|---|---|---|---|
| Uniform | 1.683 | 4.24e-3 | 1.00 | 4.61M |
| $\log(1+\min(N_k, N_{k'}))$ | 1.651 | 3.18e-3 | 1.71 | 3.17M |
| $\sqrt{N_k N_{k'}}$ | 1.706 | 2.28e-3 | 3.55 | 1,246 |
| $\min(N_k, N_{k'})$ | 1.712 | 2.05e-3 | 4.41 | 2,526 |
| **Harmonic** | 1.710 | 2.02e-3 | **4.53** | 1,849 |
| Harmonic, cap@99th pct | 1.606 | 2.40e-3 | 2.83 | 765K |

functionals of cell-specific ratios. Specifically, the population limit of $\hat{R}$ under any weight $\omega$ is

$$R_\omega \;=\; \frac{\sum_i \omega_i\, \theta_k^i}{\sum_i \omega_i\, \theta_{k'}^i},$$

which equals $p_k/p_{k'}$ only when $\theta_k^i/\theta_{k'}^i$ is the same constant across cells—i.e. when PBM holds exactly. Under mis-specification each scheme thus targets a different weighted average of the cell-specific ratios $\rho_i = \theta_k^i/\theta_{k'}^i$ (uniform $\to$ 1.68, harmonic $\to$ 1.71, $\theta$-aware oracle $\to$ 2.34 on KDD). The plug-in $\widetilde{V}$ comparison is a precision statement at fixed $\omega$—a model-based efficiency comparison—and does *not* claim that one $R_\omega$ is the "correct" position-bias ratio. Choosing $\omega$ thus involves a precision/estimand trade-off; harmonic's attractive property is that, among count-only schemes, it concentrates on high-information cells (where $\rho_i$ is estimated most precisely) without depending on noisy plug-in $\hat{\theta}_i$.

*Quantifying PBM mis-specification (Cochran's Q).* PBM is a known approximation, not a literal model – prior work [6, 14] has long established that real logs violate it. The relevant question is how *large* the deviation is. Cochran's $Q$ on the 16,132 high-information pairs ($\geq$ 5 clicks at each of positions 1 and 2) gives $Q = 6.17{\times}10^4$, df = 16,131, $Q$/df = 3.83: per-cell odds-ratio variability is $\sim$ 3.8× what binomial sampling alone would produce, well outside Poisson dispersion. This magnitude is the quantitative driver of the estimand drift discussed in Paragraph 4; plug-in $\widetilde{V}$ keeps the efficiency comparison well-posed by working at fixed $\omega$.

## 5 Synthetic and Yahoo-LTR Validation

To validate the ARE prediction in settings with ground truth, we run two controlled experiments.

*Anchor-item imbalance sweep (synthetic).* Following [1]: $M$=10 positions, $p_k = k^{-\eta}$ with $\eta = 1$, AdjacentChain estimator. A fraction $s$ of documents are "anchors" with heavy traffic at one position ($N \sim \mathrm{Unif}(50\rho, 200\rho)$) and light traffic at the other ($N \sim \mathrm{Unif}(1, 51)$); the remaining $1 - s$ are balanced ($N \sim \mathrm{Unif}(25, 75)$ at both positions). Here $s$ is the traffic split and $\rho$ is the imbalance ratio (disjoint from $r_i$ above); $s \in \{0.5, \ldots, 0.95\}$, $\rho \in \{1, \ldots, 100\}$, 500 trials per cell. At $s = 0.80$, $\rho = 100$ harmonic reduces variance by 43.6% vs. uniform and 5% vs. min; reduction grows to $\sim$ 54% at $s = 0.90, \rho = 50$. Across the full $(s, \rho)$ grid the data-equivalence factor $\mathrm{DEF}(s, \rho) = b_{\mathrm{uni}}(s, \rho)/b_{\mathrm{harm}}(s, \rho)$ – the ratio of base impressions per document at which each scheme first hits a fixed MSE threshold – has median 1.31, mean 1.30, max 1.94. Highly imbalanced cells achieve up to 2× data savings; near-balanced cells ($\rho = 1$) saturate at DEF $\approx$ 1.

**Table 5: MSE decomposition under three explicit PBM violations (anchor imbalance 50:1, split 80/20, 5,000 trials). Count-only schemes share comparable bias$^2$ within each model; harmonic achieves the lowest variance, hence the lowest MSE.**

| Model | Scheme | MSE | Variance | Bias$^2$ |
|---|---|---|---|---|
| Trust | Uniform | 0.00487 | 0.00486 | 1.0e-5 |
| | Min | 0.00298 | 0.00297 | 2.0e-5 |
| | Harmonic | **0.00272** | 0.00272 | 0.0 |
| Cascade | Uniform | 0.01376 | 0.00276 | 0.01100 |
| | Min | 0.01242 | 0.00171 | 0.01071 |
| | Harmonic | **0.01252** | 0.00147 | 0.01105 |
| Pos.-dep. | Uniform | 0.04086 | 0.03973 | 0.00113 |
| | Min | 0.03896 | 0.03770 | 0.00127 |
| | Harmonic | **0.03724** | 0.03613 | 0.00111 |

*Bias comparability under PBM violations.* We additionally simulate three explicit PBM violation models – trust bias, cascade, and position-dependent relevance [1, 5] – at imbalance ratio 50:1, split 80/20, 5,000 trials. Across all three models the count-only schemes (uniform, min, harmonic) exhibit *comparable* bias$^2$, so harmonic's lower variance translates directly into lower MSE (Table 5). Under cascade, where history-dependent examination introduces a substantial structural mismatch (bias$^2$ $\approx$ 0.011 for every ratio-unbiased scheme), the variance edge is preserved but the practical MSE gain shrinks because bias dominates.

*Yahoo-LTR semi-synthetic.* Same simulator with graded relevance from the Yahoo Learning to Rank Challenge Set 1 test split (6,936 queries; relevance grades $g \in \{0, ..., 4\}$ mapped to $r(q, d) = (2^g - 1)/15$ following NDCG; click model $P(C = 1 \mid q, d, k) = p_k\, r(q, d)$; 200 trials). Harmonic reduces variance by $\sim$ 40% vs. uniform at imbalance $\rho = 50$ (Var $1.5{\times}10^{-4} \to 8.7{\times}10^{-5}$). With 200 trials, a delta-method approximation places the 95% CI on the variance-ratio at $[0.49, 0.71]$, so the gain is well separated from 1.0 but not tightly pinned. Aggressive clipping trades variance for bias and yields $\sim$458× higher MSE.

## 6 Related Work

Intervention harvesting was introduced by [1]; [13] jointly estimates relevance and propensities via EM. Self-normalised weighted ratio estimators are standard in importance sampling and survey statistics [3, 9, 10]; inverse-variance weighting is canonical in meta-analysis [4]. The closest off-policy analogue is SNIPS [2, 11]. PBM violations on real logs are documented in [6, 14].

*Why count-only PBM weighting is still worth studying.* Given that PBM is approximate, EM-style joint estimation of relevance and propensities [13] is a natural alternative with its own bias–variance trade-offs (initialisation sensitivity, model choice). The two approaches are complementary: a relevance-agnostic weight is the right input to an EM M-step that re-uses count statistics. Empirical EM comparison is left to the regular-paper version.

## 7 Conclusion

Harmonic weighting is the inverse-variance optimum on a relevance-agnostic count-only surrogate of the PBM delta-method variance; the actual inverse-variance solution ($\theta$-aware oracle) is plug-in infeasible at scale (ARE $=$ 0.002). On full KDD, $\mathrm{ARE}(\mathrm{harmonic}, \mathrm{uniform}) = 4.53$ on the surrogate, with our two proposals essentially tied empirically; harmonic is preferred because of Theorem 3's unconditional guarantee. Under PBM misspecification, choosing $\omega$ is a precision/estimand trade-off. Future work: shrinkage $\theta$-aware oracle; EM [13] comparison; click models beyond PBM [6, 14].

## GenAI Usage Disclosure

The authors used a large-language-model assistant for code generation (KDD aggregation, plug-in variance script, LaTeX table formatting), and for copy-editing. All experimental design, mathematical derivations (Theorems 2–4 and Eq. (2)), interpretation, citations, and final wording were authored and verified by the authors. No LLM-generated content was used without human review. Full code, scripts, and data-processing pipelines are provided as supplementary material.

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